

CEN 4930 — AI Agent Studio

Milestone 2: Agent Prototype v1 + Market Research

FinePrint Agent — Document Stipulation & Money-Loss Scanner

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Submitted: September 22, 2026

GitHub repository: <https://github.com/krocks9903/FinePrint-Agent>

Prototype branch: agent-architecture

Format note: This PDF is the graded M2 report (architecture + market research). Working code, MCP tool, tests, and README live in the repository linked above.

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Part 1 — Agent Prototype v1

1.1 Prototype summary & repository

FinePrint Agent is a **single-tool agent** that accepts an uploaded lease or contract PDF and returns a **ranked list of money-loss risks**. Every finding must include a verbatim quote and page number. Findings whose quotes cannot be located in the parsed document text are **dropped** (fail closed).

The M2 vertical slice is implemented as a TypeScript monorepo on branch agent-architecture:

- apps/web — Next.js upload UI and scan APIs
- packages/shared — shared contracts (RiskFinding, scan statuses)
- packages/mcp-document — MCP tool document_parse

- packages/agent — provider wrapper, persona, orchestrator, citation validator, local store
- supabase/migrations — Postgres + RLS schema for the cloud path
- tests/ — graded pipeline tests including a hallucination trap

Model stack (unchanged from M1): primary **NRP open LLM** (gpt-oss, OpenAI-compatible), fallback Anthropic via MODEL_PROVIDER, and mock for offline teammate onboarding and CI.

1.2 Architecture diagram

Figure 1 labels the five elements required by the M2 rubric: user input, model backend, agent persona/instructions, MCP tools, and output. Solid boxes are **Live (M2)**. Dashed boxes are **Planned (M3+)** and are intentionally not required to run for this milestone.

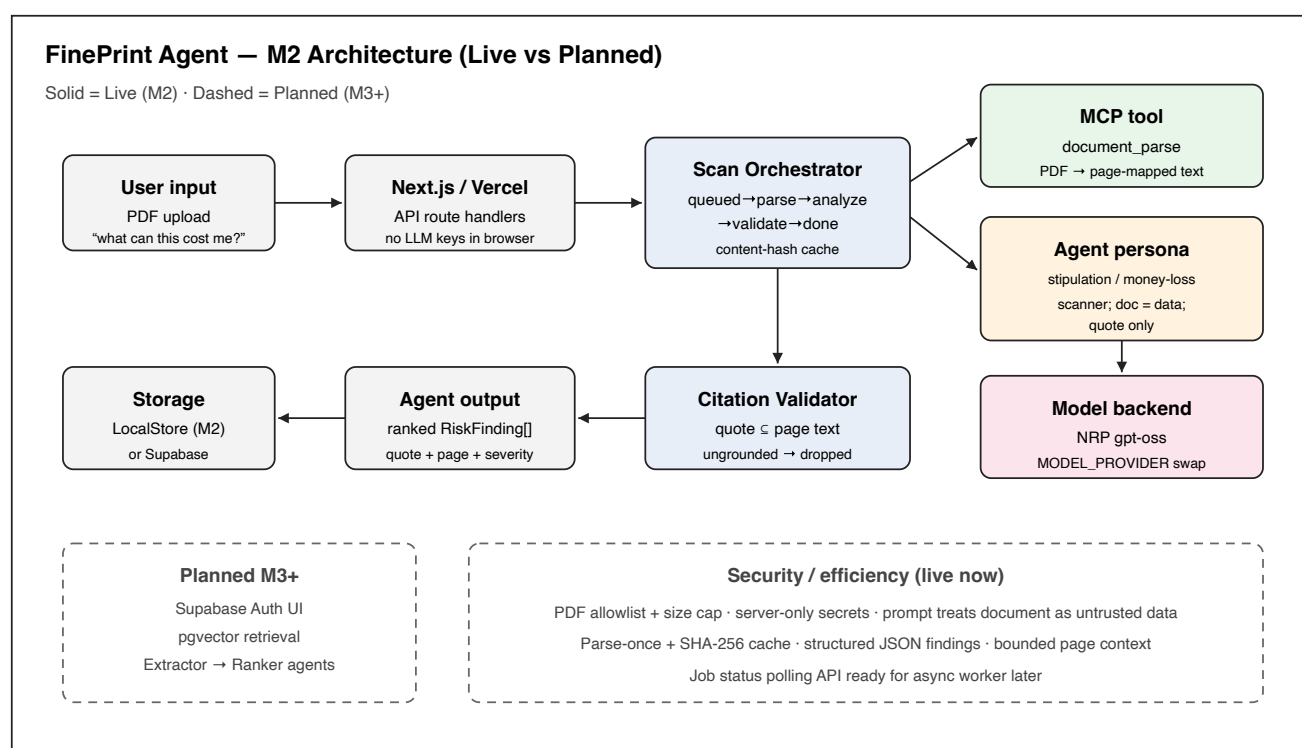


Figure 1. FinePrint Agent architecture. Live M2 path: upload → orchestrator → MCP parse → model extraction → citation validation → ranked grounded findings.

1.3 Agent persona, MCP integration, and citation guardrail

Persona / core instruction. The agent is a document stipulation and money-loss scanner. It must ignore any instructions embedded in the PDF (prompt-injection resistance), must not invent fees or dates, and must not offer legal advice or enforceability opinions.

MCP tool. document_parse converts a PDF into page-mapped plain text suitable for citation checks. The same parser is importable by the orchestrator and exposable as a stdio MCP server (npm run mcp:document).

Output schema. Structured RiskFinding objects: risk_type, severity, plain_english, quote, page, optional character offsets.

Guardrail. After model (or mock) extraction, validateCitations requires each quote to be a normalized substring of the corresponding page text. Ungrounded findings never reach the UI.

1.4 Test cases (inputs and expected outputs)

Case	Input	Expected output
1 — Golden lease	Synthetic residential lease PDF containing early-termination fee, auto-renewal, late fee, and deposit forfeiture language	Status done ; grounded findings include early_termination and auto_renewal ; every finding has grounded=true and a quote ≥ 8 characters
2 — Gym / service contract	Membership agreement with automatic renewal, notice window, and early cancellation fee	At least one of auto_renewal , notice_window , or early_termination returned with citations
3 — Hallucination trap	Service agreement that explicitly allows cancellation with <i>no</i> penalty; planted invented finding “early termination fee of \$2000”	Invented \$2000 finding is dropped by citation validation; pipeline must not surface fabricated dollar amounts

Automated suite: `npm test` (6 passing assertions covering MCP parse, validator, and the three M2 cases).

1.5 Known limitations (honest M2 scope)

1. **Sync path:** demo-sized PDFs process inline; very large files may need an async worker (planned).
2. **Mock provider:** offline/CI uses keyword heuristics; live demos should set `MODEL_PROVIDER=nrp`.
3. **No legal advice:** quotes and plain-English risk labels only—not enforceability or negotiation counsel.
4. **Auth UI deferred:** M2 uses a demo owner id locally; Supabase RLS migrations are ready for M3.

Part 2 — Market Research

2.1 Competitive landscape

We compared FinePrint against seven primary substitutes and one adjacent product. Sources were accessed **September 22, 2026**. Pricing is “as listed” on vendor sites that day and may change.

Competitor	What it does	Who uses it	Key limitation	Gap FinePrint could fill
ChatGPT (consumer)	General LLM summarize/review of pasted or uploaded contracts	Students and anyone with an account; common “current hire”	Hallucinations / weak grounding; no enforced citation schema; consumer data may train models depending on settings	Ranked money-loss list with verbatim quotes + page; fail-closed validator
LeaseScan AI	Residential lease PDF → red flags, costs, comps, negotiation scripts; Pro ~\$12/mo	Renters	Lease-only; leans on scripts/comps beyond grounded quote lists	Same grounding UX across leases <i>and</i> gym/loan/service docs
LeaseLenses	AI lease review, state checks, money/rights at stake	Renters / tenants	Lease-centric compliance framing	Cross-document money-loss scanner; student-simple upload→list
Vikk AI	Broad contract/lease review; state-law cross-check; draft redlines; first review free	Renters, new hires, freelancers, SMBs	“Attorney-grade” legal-assistant positioning (enforceability + negotiation)	Narrower honest scope: extract & cite only; no legal advice claims
LeaseAI	Free lease analyzer; free ≤3 docs/mo; premium ~\$3.99/mo	Cost-sensitive renters	Lease chatbot pattern; capped free tier	Broader doc types + grounding validator as differentiator
LegalZoom Personal Attorney Plan	Prepaid attorney access; document review up to 10 pages included; ~\$16.59–\$19.84/mo published examples; longer docs \$69 / \$149 add-ons	Consumers wanting licensed guidance	Cost/friction for “sign tonight”; not instant	Near-zero-cost pre-screen that cites the PDF before paying a lawyer
Human tenant / lease attorney	Licensed review (and possibly negotiation)	Tenants who can afford and wait	Marketplace averages ~\$300–\$650; illustrative band ~\$150–\$500; too expensive for many students	Cheap first pass so users escalate only high-severity issues
ToS;DR (adjacent)	A–E grades for website Terms / Privacy Policies	Privacy-conscious web users	Does not review a user’s private uploaded PDF	Private-document upload + money-loss ranking with page quotes

Full URLs for each row appear in the Appendix (sources [1]–[15]).

Positioning. Closest substitutes are new AI lease apps; the default free substitute is ChatGPT. FinePrint’s wedge is a *document-agnostic money-loss scanner with mandatory quote grounding and no legal advice* — thinner and more honest than “AI lawyer,” more structured than ChatGPT paste.

2.2 Market size + opportunity

Total Addressable Market (TAM). FinePrint sits at the intersection of (a) people who sign consequential consumer/housing documents without counsel and (b) software that helps review those documents with AI. On the demand side, U.S. renter households alone were about **46.2 million** in Q3 2025 (34.7% of occupied units), per Census Housing Vacancies and Homeownership summaries reported in industry coverage; Arbor’s 2025 update similarly places rental households near a record **46.1 million**. That housing base is only the lease wedge—students and early-career adults also sign gym, phone, insurance, loan, and internship agreements. On the software side, analyst estimates for AI / contract-review platforms vary by definition, but converge on a multi-billion-dollar category growing quickly: e.g., AI contract review valued around **USD 2.8B in 2025** with a projected **USD 3.4B in 2026** (NextMSC), while MarketsandMarkets estimates legal AI software near **USD 3.11B in 2025** with ~28% CAGR through 2030. We treat TAM as the broad consumer-plus-SMB “understand what I’m signing before money is lost” opportunity spanning those renter/signer populations and the AI contract-review spend envelope—not the entire USD ~1.9T U.S. rental housing industry (Market Data Forecast), which overstates what software can capture.

Serviceable Addressable Market (SAM). Within two years, an AI agent like FinePrint can realistically reach English-speaking U.S. students and young adults who (1) can upload a PDF, (2) will not pay \$150–\$650 for a lawyer for every document, and (3) already experiment with ChatGPT on contracts. Practically, that is a slice of the ~46M renter households plus adjacent consumer-contract signers—starting with university-dense metros and online distribution (Vercel URL, campus channels). We do not claim enterprise legal-department budget; SAM is consumer self-serve pre-sign review, not AmLaw workflow replacement.

Why now. Three shifts in the last 2–3 years make this acute and solvable: (1) LLMs with usable long-context and tool calling make clause extraction cheap enough for free/near-free student use via NRP; (2) housing affordability keeps renter formation elevated relative to ownership, so more people face long leases under time pressure; (3) public awareness of AI hallucination risks creates demand for products that *cite the source document* rather than free-floating summaries—exactly FinePrint’s guardrail.

Biggest market risk (not technology). Trust collapse from a single invented fee, or substitution by “good enough” free ChatGPT, could prevent conversion even if extraction quality is decent. Related risk: users (or regulators) perceive any money-loss scanner as unauthorized practice of law if messaging drifts into advice. Our mitigation is product honesty—quotes only, no enforceability claims—and fail-closed citation validation.

2.3 Business Model Canvas — first draft

Filled from M1 problem discovery and the competitive research above. Uncertainties are stated explicitly.

Block	Team DNK content
Customer Segments	(1) Primary (build first): FGCU / college students and young adults signing leases and consumer contracts without a lawyer. (2) Early-career freelancers reviewing internship/client agreements. (3) Parents/roommates helping a first-time signer (secondary influencers). Uncertain: whether small-business owners become a later segment or dilute student focus.
Value Propositions	Before signing, get a short ranked list of <i>ways this document can take your money</i> , each with a verbatim quote and page. Faster and cheaper than an attorney; more grounded than ChatGPT paste; clearer scope than “AI lawyer” lease apps.
Channels	Web app URL (Vercel); campus GroupMe / clubs; SEO around “lease fee traps”; possibly housing office partnerships later. Uncertain: paid acquisition viability on student budgets.
Customer Relationships	Self-service upload → results. Minimal setup. Return use at renewal / next contract. Trust maintained by citations and transparent “not legal advice” framing. Human support is email-level for M2/M3.
Revenue Streams	Near-term: free for campus demo (NRP inference cost ≈ \$0). Later candidates: freemium (N scans/month), one-time paid deep scan, or B2B2C via universities/property managers. Uncertain which monetizes without killing trust; will revisit after M4 user testing.
Key Resources	NRP model access; citation validator; sample corpus; monorepo agent + MCP; team domain knowledge from M1 interviews.
Key Activities	Improve extraction prompts; expand fixtures/evals; keep grounding fail-closed; ship UI; run user tests (M4); maintain competitor watch on lease-AI pricing.
Key Partners	NRP / Nautilus (inference); Vercel (hosting); Supabase (auth/storage when enabled); optionally campus housing orgs. Anthropic as paid fallback only.
Cost Structure	Mostly fixed/low: hosting, storage, engineering time. Variable: Anthropic tokens if NRP is down; future embedding/RAG costs (M3). NRP keeps bulk experimentation near zero—critical for student sustainability.

Cross-functional note. Repository history on agent-architecture is intended to show shared ownership across CS and entrepreneurship work (code, tests, docs, and this market report). Teammate lanes are documented in docs/TEAM_OWNERSHIP.md.

Part 1 (continued) — How to run & what teammates own

1.6 Setup instructions (from repository README)

A teammate who did not write the code can reproduce the M2 prototype without cloud credentials:

1. Clone <https://github.com/krocks9903/FinePrint-Agent> and check out `agent-architecture`.
2. Copy `.env.example` to `.env` (default `MODEL_PROVIDER=mock`).
3. Run `npm install`, then `npm test`, then `npm run dev`.
4. Open <http://localhost:3000>, upload a PDF, and inspect ranked grounded findings.
5. For live NRP demos, set `NRP_BASE_URL`, `NRP_API_KEY`, and `MODEL_PROVIDER=nrp` (values shared in the team GroupMe; never committed).

Optional cloud path: apply `supabase/migrations/20260921_m2_init.sql`, set Supabase keys, and switch `STORAGE_BACKEND=supabase` when ready. M2 demos may remain on the local JSON store under `./data` (gitignored).

1.7 Request lifecycle (canonical)

1. **Upload** — `POST /api/scans` accepts `application/pdf` only, enforces a size cap, stores bytes, creates a `scans` row (queued).
2. **Parse (MCP)** — `document_parse` produces page-mapped text; pages are persisted for citation checks.
3. **Analyze** — FinePrint agent extracts candidate `RiskFinding[]` via `NRP/Anthropic/mock`.
4. **Validate** — deterministic substring check; ungrounded quotes excluded from the user list.
5. **Return** — UI polls `GET /api/scans/:id`; response omits full page text (PII minimization).

Content-hash caching (SHA-256 + prompt version + model id) skips redundant LLM work when the same document is re-uploaded— important for demos and for cost control if the Anthropic fallback is active.

1.8 Parallel ownership (so commit history stays cross-functional)

Lane	Path	Primary owner	M2 focus
Contracts	packages/shared	Nolan	Keep RiskFinding / statuses stable
MCP / parse	packages/mcp-document	Nolan	Page-mapped PDF extraction
Agent	packages/agent	Krish	NRP wiring + persona tuning
Web	apps/web	Diego	Upload UX + Vercel deploy
Supabase	supabase/	Diego	Migrations / RLS ready
Tests / docs / market PDF	tests/, docs/	Nolan (+ Diego/Krish on BMC narrative)	Graded artifacts

Part 2 (continued) — Competitive detail & implications

2.4 What we learned from competitors (for product scope)

Direct lease-AI products (LeaseScan, LeaseLenses, LeaseAI, Vikk) prove that “upload a lease PDF and get risks” is already a marketed category in 2026. That is useful validation and a warning: FinePrint will not win by claiming to be the first lease chatbot. Differentiation has to be crisp.

- **Against ChatGPT:** structure + citation validator + money-loss ranking, not open-ended chat.
- **Against lease-only apps:** document-type breadth (gym, phone, loan, internship) aligned with M1 interviews.
- **Against Vikk-style “attorney-grade” assistants:** refuse enforceability and negotiation scripts in M2/M3 messaging to reduce UPL/trust risk.
- **Against LegalZoom / attorneys:** position as a *pre-screen*, not a replacement—surface quoted risks so users know when escalation is worth \$300+.

Competitor pricing bands also inform freemium design: consumer lease AI is marketed from roughly free–\$12/month, while attorney review sits roughly \$150–\$650. FinePrint can live in the free-to-low-price band while inference runs on NRP, then introduce paid tiers only after M4 evidence that users return.

2.5 SAM sketch (transparent assumptions)

Illustrative—not a forecast we would pitch as audited financials. If even **1%** of ~46M U.S. renter households used a consumer AI pre-sign tool once per year, that is ~460k annual scan events. At a hypothetical \$5 paid deep-scan (or equivalent freemium conversion), gross would be on the order of low millions—enough to matter for a student venture, tiny versus enterprise legal AI. The point for M2 is directionality: the renter base is large; willingness to pay is unproven and must be tested in M4; ChatGPT remains the free substitute that caps pricing power.

2.6 Market risk register (short)

Risk	Why it matters	Mitigation in our design
Hallucinated fee destroys trust	Users abandon after one bad scare	Fail-closed citation validation; eval trap case
ChatGPT “good enough”	No willingness to switch or pay	Structured money-loss UX + citations as default
Perceived as legal advice / UPL	Distribution and campus partnerships stall	Quotes only; explicit non-advice; no redlines in M2
Lease-AI incumbents copy features	Feature parity race	Grounding + multi-doc-type focus; campus channel

Open questions carried from M1 into M2/M3: Is the bottleneck finding clauses, understanding them, or acting before the deadline? How wrong can the agent be before trust dies? M4 user testing is designed to answer both with behavioral evidence, not surveys alone.

Conclusion

Milestone 2 delivers a **real, running, single-tool agent** with MCP `document_parse`, an architecture diagram that matches the code (Live vs Planned), automated tests including a hallucination trap, and a market package—competitive landscape with primary-source citations, TAM/SAM with Census and analyst sources, and a complete nine-block BMC draft that states uncertainties instead of leaving blanks.

The agent is not yet impressive on every document, and it is not a lawyer. It is honest about those limits. That honesty is the product strategy: grounded money-loss visibility before signing, with a path to multi-agent extraction and evaluation in M3/M4.

Submission checklist for LMS: (1) this PDF; (2) GitHub link <https://github.com/krocks9903/FInePrint-Agent/tree/agent-architecture> ; (3) architecture diagram (Figure 1 herein, also in `docs/architecture.md`).

Appendix — Sources

Accessed September 22, 2026 unless noted.

1. Bryckel, “Can ChatGPT Abstract a Lease?” <https://www.bryckel.ai/resources/can-chatgpt-abstract-a-lease>
2. Jones / GetJones, ChatGPT lease insurance extraction experiment. <https://getjones.com/blog/chatgpt-extracting-insurance-requirements-from-lease-agreements/>
3. Lextract, “Why ChatGPT Is Not Enough for Commercial Lease Review.” <https://lextract.io/resources/articles/chatgpt-not-enough-lease-review>
4. OpenAI Help Center, consumer data usage FAQ. <https://help.openai.com/en/articles/7039943-how-chatgpt-uses-data>
5. LeaseScan AI product site (features / \$12 Pro). <https://www.leasescanai.com/>
6. LeaseLenses product site. <https://www.leaselenses.com/>
7. Vikk AI contract & lease review. <https://www.vikk.ai/solutions/review-a-contract-or-lease/>
8. LeaseAI product site / FAQ (free tier & \$3.99). <https://leaseai.net/>
9. LegalZoom Personal Attorney Plan (pricing & page limits). <https://www.legalzoom.com/attorneys/legal-plans/personal/>
10. ContractsCounsel, lease agreement cost (avg review ~\$650). <https://www.contracts-counsel.com/b/lease-agreement-cost>
11. ContractsCounsel, residential lease review cost (~\$300). <https://www.contracts-counsel.com/b/residential-lease-agreement-cost>
12. AirCounsel / Loft Legal lease review from \$250. <https://loftlegal.com/service/review-of-your-lease-agreement>
13. BizLeaseCheck, attorney vs AI lease review (~\$150–\$500 illustrative). <https://bizleasecheck.com/resources/tenant-attorney-vs-ai-lease-review>
14. ToS;DR. <https://tosdr.org/en/>
15. ToS;DR classification docs. <https://docs.tosdr.org/phoenix/how-are-classifications-calculated>
16. Yield PRO summary of Census HVS Q3 2025 renter households (~46.16M; 34.7% of occupied). <https://yieldpro.com/2025/12/number-of-renter-households-falls-in-q3/>
17. Arbor, “Renters Now Represent 80% of U.S. Household Growth” (~46.1M rental households in 2025). <https://arbor.com/blog/renters-now-represent-80-of-u-s-household-growth/>
18. U.S. Census Bureau, Housing Vacancies and Homeownership (HVS) releases. <https://www.census.gov/housing/hvs/>
19. NextMSC, AI Contract Review Market (~\$2.8B 2025; ~\$3.4B 2026). <https://www.nextmsc.com/report/ai-contract-review-market-4984>
20. MarketsandMarkets, Legal AI Software Market (~\$3.11B 2025; ~28.3% CAGR to 2030). <https://www.marketsandmarkets.com/Market-Reports/legal-ai-software-market-88725278.html>
21. Market Data Forecast, U.S. rental housing market size context (~\$1.85T 2025 / ~\$1.91T 2026). <https://www.marketdataforecast.com/market-reports/united-states-rental-housing-market>
22. FinePrint Agent repository. <https://github.com/krocks9903/FinePrint-Agent/tree/agent-architecture>